

The empirical recurrence for OEIS A234783: recovering a conjecture that is not written down

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Abstract

OEIS A234783 records “Empirical recurrence of order 70”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 71$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 70, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 70 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A234783 is “Number of $(n+1) \times (5+1)$ 0..3 arrays with no adjacent elements equal and with each 2×2 subblock having the number of clockwise edge increases equal to the number of counterclockwise edge increases.”. It has offset 1 and begins

125228, 17521780, 2502596188, 359793857812, 51845281856108, 7476894104333052, 1078605317680077396, 15561

The entry states:

Empirical recurrence of order 70 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 08:26:43, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 973$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 71$ of the 973, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 71$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 1946$ exact terms of the model returns order 70 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{70} c_i a(n-i),$$

c_1	324	c_{19}	5139427984716424810557	c_{37}	−4415531565390270910566613371
c_2	−39638	c_{20}	−82029696703016595502402	c_{38}	3085987321444282689953699174
c_3	2563357	c_{21}	302943529549060965986102	c_{39}	916894807671605192280872310
c_4	−99678028	c_{22}	−320615792113076602489829	c_{40}	−3639508264916860485380515902
c_5	2422898292	c_{23}	−1657115306226758776142290	c_{41}	2970759642505242646432741014
c_6	−34567057037	c_{24}	8068597001512785547875111	c_{42}	−466131808362506395905349106
c_7	163766577224	c_{25}	−13376217033818619829173627	c_{43}	−1230709325386156350054319773
c_8	3953924057449	c_{26}	−9007290254306026204266458	c_{44}	1255759127222850231908023237
c_9	−93675129477373	c_{27}	87525051383286874503714054	c_{45}	−505110331437242495850314616
c_{10}	854871679192289	c_{28}	−162580189876993465142148189	c_{46}	−31481521623588192528299729
c_{11}	−1010373911023728	c_{29}	23526211232093279356308963	c_{47}	140514212979011798478024679
c_{12}	−64538210724303463	c_{30}	471208575610332879083549054	c_{48}	−65945478796808695137450941
c_{13}	764374153831286079	c_{31}	−918944556261933248058181605	c_{49}	5726585916010713017059660
c_{14}	−3535065266008603304	c_{32}	402135296399868083040981818	c_{50}	7470898741842649706262107
c_{15}	−6650643116239888763	c_{33}	1334981348496023848871530178	c_{51}	−3412956298232072922390919
c_{16}	199768366558925441548	c_{34}	−2739540886500718334150601911	c_{52}	262494321509038838681984
c_{17}	−1284148298178750659633	c_{35}	1609967371243683762919217023	c_{53}	252280220540757965905947
c_{18}	3495370310405908549678	c_{36}	1882070046189479230490753891	c_{54}	−85247779462303361462547

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 70, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $70 + 4$ terms removed returns order 70 as well, so $\deg q_{\min} = 70 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 11 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $70 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 70 that this sequence satisfies — and that family turns out to have exactly one member.

References

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